

## Use cases for Internet of Things

Grafana's ability to process and visualise real-time, time-series data from a variety of sources makes it a great tool for Internet of Things (IoT) applications. Some typical use cases are listed below.

### **Environmental monitoring:**

Grafana tools help monitor environmental factors like temperature, humidity, air quality, and noise levels by gathering and displaying data from Internet of Things sensors. Data sources like MQTT, AWS IoT, or custom APIs are vital for such use cases. Grafana helps to generate dynamic and powerful visualisations that display environmental data through a dynamic dashboard.

**Soil monitoring:** One of the most important uses of IoT in environmental sciences and agriculture is soil monitoring. Farmers and researchers can make well-informed decisions to maximise crop production, conserve water, and preserve soil health by visualising soil data in Grafana. Grafana dashboards display soil moisture, temperature, and nutrient levels. The tool also analyses trends for planning irrigation.

### **Predictive maintenance:**

Predictive maintenance anticipates and fixes equipment issues before they happen by using data analytics. Organisations can optimise maintenance schedules, minimise downtime, and monitor equipment health by integrating IoT sensors, data storage, and Grafana dashboards.

**Smart energy management:** By facilitating real-time monitoring, analysis, and optimisation of energy production and consumption, Grafana can be a potent tool for smart energy management. Organisations may monitor energy consumption, identify inefficiencies, and make data-driven decisions to lower expenses and enhance sustainability by combining Grafana with IoT devices and data sources.

**Patient monitoring:** Grafana is a great tool for patient monitoring since it lets medical professionals see, evaluate, and notify patients in real time. Clinicians may improve patient outcomes by continually monitoring vital signs, identifying abnormalities, and promptly intervening by integrating Grafana with IoT medical equipment.

Table 1: Comparison of Grafana with other popular tools

| Salient parameter                | Grafana                                                                                      | ThingsBoard                                                                                                             | Kibana                                             | Freeboard                               | Node-RED                                    |
|----------------------------------|----------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------|-----------------------------------------|---------------------------------------------|
| <b>Setup</b>                     | Easy setup for basic use. However, it requires manual configuration for some advanced use    | Easy                                                                                                                    | Similar to Grafana but optimised for Elasticsearch | Easy                                    | Easy                                        |
| <b>Visualisation capacity</b>    | Provides extensive real-time, time-series data visualisation. Highly customisable dashboards | IoT platform with a focus on device management and telemetry. Predefined widgets and templates; moderately customisable | Limited but powerful visualisation                 | Limited visualisation capacity          | Limited visualisation capacity              |
| <b>Data source support</b>       | Wide range of data support                                                                   | IoT-centric data support                                                                                                | Limited data support                               | Data support is limited to JSON and API | Moderate data support                       |
| <b>Real-time data capability</b> | Yes                                                                                          | Yes                                                                                                                     | Yes                                                | Yes                                     | Yes                                         |
| <b>Extensibility</b>             | High; extensibility can be achieved using plugins                                            | Moderate extensibility                                                                                                  | Moderate extensibility                             | Low                                     | High -- can be achieved using custom nodes. |
| <b>Performance</b>               | Excellent for IoT, particularly for time series applications                                 | Good for IoT applications                                                                                               | Excellent for applications having logs data        | Moderate                                | Moderate                                    |
| <b>Integration capability</b>    | Excellent                                                                                    | Built-in IoT integration                                                                                                | Integration is good with ELK                       | Limited                                 | Flexible                                    |